

# Suna Choe

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## Education

**Sungkyunkwan University, Korea**  
Bachelor of Computer Science & Engineering

**Mar. 2022 – Feb. 2027 (Expected)**

**Zurich University of Applied Sciences, Switzerland**

**Feb. 2024. – Jul. 2024**

## Experiences

**Automation Lab, Sungkyunkwan University**

**Suwon-si, Gyeonggi-do**

Research Intern, Multi-Camera VIO System & Calibration Analysis

Dec. 2025 – Jan. 2026

- Prototyped a stereo VIO system using **ROS2** and **BASALT** with Fisheye cameras.
- Investigated extrinsic calibration instability, verifying the correlation between AprilGrid dimensions and IMU-Camera trajectory estimation errors.
- Developed a data post-processing tool to synchronize asynchronous sensor topics, improving the reliability of the odometry initialization process.

**Sehwa, Industry-University Cooperation**

**Yuseong-gu, Daejeon Metropolitan City**

Software Engineer Intern, Railway Simulation Team

Apr. 2025 – Dec. 2025

- **Analyzed a large-scale Rust codebase** to document core interaction flows, reducing onboarding time and ensuring UI consistency.
- Redesigned object structures to **resolve critical model-reference bugs** affecting interlocking logic, significantly improving simulation stability.
- **Standardized a Docker-based Linux development environment** using Xpra, ensuring consistency across local and remote servers.

## Awards & Certifications

**The 3rd SKKU Autonomous Driving AI Chip Design Competition**

**Jun. 2025 – Jul. 2025**

- **Enhanced Control Stability:** Mitigated abrupt steering fluctuations and stabilized cornering by implementing Kanayama control and soft-steering algorithms.
- **Data-Driven Optimization:** Conducted iterative track testing to analyze the impact of steering sensitivity, tolerances, and speed, establishing a **systematic parameter tuning process**.

## Projects

**xv6-riscv OS | Course**

**Mar. 2025 – Jun. 2025**

Tech Stack: C

- Developed the **EEVDF (Earliest Eligible Virtual Deadline First)** scheduler by calculating vruntime, vdeadline, and eligibility to ensure both fairness and latency requirements.
- Implemented **virtual memory management** with a **page fault handler** and support for mmap/munmap (File & Anonymous mapping).
- Page Replacement & Swapping using **LRU (Least Recently Used)** list with a **Clock algorithm** to optimize physical memory utilization under high demand.

**SoleMatch | Plantar Analysis & Shoe Recommendation App**

**Sep. 2024– Nov. 2024**

Tech Stack: Python, React Native, Arduino

- Developed an **end-to-end solution** integrating Arduino pressure sensors with serial communication to capture and visualize foot pressure distribution data.
- Implemented **computer vision-based foot measurement** using **OpenCV** library to measure foot length and width from camera images.
- Designed and built **RESTful APIs** with Flask and SQLite for user data management, image processing, and sensor data integration.

**CodEco | Carbon Footprint Analysis & Code Optimization Platform | Course**

**Mar. 2023 – Jun. 2023**

Tech Stack: Next.js

- **Formalized user requirements** by applying **Use Case Modeling** and adhering to IEEE Std 830, crystallizing interface requirements within the **Software Requirements Specification (SRS)**.
- Engineered a modular frontend architecture based on **Component-Based Design (CBD)** principles and authored the **Software Design Specification (SDS)** following IEEE Std 1016.
- **Visualized asynchronous data flows** using **UML Sequence Diagrams** to define clear communication specifications between the client and server.

## Technical Skills

C/C++, Python, Scikit-Learn, Pandas, Rust, ROS2, Linux, Git, Docker, MySQL, TypeScript, JavaScript, HTML/CSS, React, React Native, Next.js, TailwindCSS